

3NF Decomposition Example

Consider a relation R with attributes $\mathcal{U} = \{A, B, C, D, E, G, H, K\}$ and the set of functional dependencies

$$\mathcal{F} = \{ABH \rightarrow CK, A \rightarrow D, C \rightarrow E, BGH \rightarrow L, L \rightarrow AD, E \rightarrow L, BH \rightarrow E\}$$

a) Compute

- $(A)^+ =$ $(ABH)^+ =$
- $(C)^+ =$ $(BGH)^+ =$
- $(L)^+ =$ $(E)^+ =$
- $(BH)^+ =$

b) Find all the candidate keys of the relation R .

c) Find a minimal cover of the relation R .

- **Step 1** *Decompose all FDs of the form $X \rightarrow A$, where A is a single attribute.*

$$\mathcal{G} = \{$$

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- **Step 2** *Eliminate redundant attributes from LHS.*

Algorithm:

If FD $XB \rightarrow A \in \mathcal{F}$ (where B is a single attribute) and $X \rightarrow A$ is entailed by $\mathcal{F}$, then B was unnecessary

- Can an attribute be eliminated from $ABH \rightarrow C$?

$$(AB)^+ =$$

$$(AH)^+ =$$

$$(BH)^+ =$$

Answer:

- Can an attribute be eliminated from $ABH \rightarrow K$?

$$(AB)^+ =$$

$$(AH)^+ =$$

$$(BH)^+ =$$

Answer:

- Can an attribute be eliminated from $BGH \rightarrow L$?

$$(BG)^+ =$$

$$(BH)^+ =$$

$$(GH)^+ =$$

Answer:

- Can an attribute be eliminated from $BH \rightarrow E$?

$$(B)^+ =$$

$$(H)^+ =$$

Answer:

$$\mathcal{G} = \{$$

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- **Step 3** Delete redundant FDs from $\mathcal{G}$.

Algorithm:

If $\mathcal{G} - \{f\}$ entails f , then f is redundant: If f is $X \rightarrow A$ then check if $A \in X_{\mathcal{G}-\{f\}}^+$

- Can $BH \rightarrow C$ be eliminated from $\mathcal{G}$?

$$(BH)_{\mathcal{G}-\{BH \rightarrow C\}}^+ =$$

- Can $BH \rightarrow K$ be eliminated from $\mathcal{G}$?

$$(BH)_{\mathcal{G}-\{BH \rightarrow K\}}^+ =$$

- Can $A \rightarrow D$ be eliminated from $\mathcal{G}$?

$$(A)_{\mathcal{G}-\{A \rightarrow D\}}^+ =$$

- Can $C \rightarrow E$ be eliminated from $\mathcal{G}$?

$$(C)_{\mathcal{G}-\{C \rightarrow E\}}^+ =$$

- Can $BH \rightarrow L$ be eliminated from $\mathcal{G}$?

$$(BH)_{\mathcal{G}-\{BH \rightarrow L\}}^+ =$$

- Can $L \rightarrow A$ be eliminated from $\mathcal{G}$?

$$(L)_{\mathcal{G}-\{L \rightarrow A\}}^+ =$$

- Can $L \rightarrow D$ be eliminated from $\mathcal{G}$?

$$(L)_{\mathcal{G}-\{L \rightarrow D\}}^+ =$$

- Can $E \rightarrow L$ be eliminated from $\mathcal{G}$?

$$(E)_{\mathcal{G}-\{E \rightarrow L\}}^+ =$$

- Can $BH \rightarrow E$ be eliminated from $\mathcal{G}$?

$$(BH)_{\mathcal{G}-\{BH \rightarrow E\}}^+ =$$

$\mathcal{G} = \{$

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d) Produce a 3NF decomposition of the relation R .

- **Step 1** Compute a minimal cover $\mathcal{G}$.

$\mathcal{G} = \{$

- **Step 2** Partition $\mathcal{G}$ into sets $\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n$ such that the LHS of all elements of $\mathcal{G}_i$ are the same.

- **Step 3** For each $\mathcal{G}_i$ form schema $R_i = (\mathcal{U}_i, \mathcal{G}_i)$, where $\mathcal{U}_i$ is the set of all attributes mentioned in $\mathcal{G}_i$.

- **Step 4** If no R_i is a superkey of R , add schema $R_0 = (\mathcal{U}_0, \{\})$, where $\mathcal{U}_0$ is a key of R .